Simultaneous Equations & Inequalities

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The temperature θ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where t is time in minutes and k is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of t when $\theta = 30$.

Separate variables and integrate: $\int \frac{1}{\theta - 20}\, d\theta = -\int k\, dt \Rightarrow \ln(\theta - 20) = -kt + C$ Apply initial condition $t = 0, \theta = 80$: $\ln(60) = C \Rightarrow \ln\left(\frac{\theta - 20}{60}\right) = -kt \Rightarrow \theta - 20 = 60e^{-kt}$ Apply condition $t = 10, \theta = 50$: $30 = 60e^{-10k} \Rightarrow e^{-10k} = \frac{1}{2} \Rightarrow k = \frac{\ln 2}{10} \approx 0.06931$ Set $\theta = 30$.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Differential Equations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Solve the separable DE $\frac{dy}{dx} = \frac{x + 1}{y}$ given $y = 2$ when $x = 0$.

Separate: $y\, dy = (x + 1)\, dx$. Integrate: $\frac{y^2}{2} = \frac{x^2}{2} + x + c$. At $(0, 2)$: $2 = c$. So $y^2 = x^2 + 2x + 4$.

4 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$.

1. Base Step ($n = 1$): LHS $= 2(1) - 1 = 1$, RHS $= 1^2 = 1$. Since LHS = RHS, the base case $P(1)$ is true. 2. Inductive Hypothesis: Assume $P(k)$ is true for some integer $k \geq 1$: $1 + 3 + 5 + \ldots + (2k - 1) = k^2$. 3. Inductive Step ($n = k + 1$): Show that $1 + \ldots + (2k - 1) + (2(k + 1) - 1) = (k + 1)^2$: LHS $= [1 + 3 + \ldots + (2k - 1)] + (2k + 1)$. By the inductive hypothesis, substitute k^2.

5 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

Substitute $\cos(2\theta) = 1 - 2\sin^2\theta$ to express the entire equation in terms of $\sin\theta$: $(1 - 2\sin^2\theta) + 3\sin\theta = 2$. $-2\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0$. Factor the quadratic in $\sin\theta$: $(2\sin\theta - 1)(\sin\theta - 1) = 0$. Case 1: $2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$. On $[0, 2\pi)$, $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$. Case 2: $\sin\theta - 1 = 0 \Rightarrow \sin\theta = 1$. On.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In a 2D Cartesian plane, points A, B, C have coordinates $A(2, -1)$, $B(5, 3)$, and $C(-1, k)$. (a) Find the vectors $\vec{AB}$ and $\vec{AC}$ in component form. (b) Find the value of k for which $\vec{AB}$ is perpendicular to $\vec{AC}$. (c) For $k = 3$, calculate the exact cosine of the angle between $\vec{AB}$ and $\vec{AC}$.

(a) $\vec{AB} = (5 - 2; 3 - (-1)) = (3; 4)$. $\vec{AC} = (-1 - 2; k - (-1)) = (-3; k + 1)$. (b) Perpendicularity condition $\vec{AB} \cdot \vec{AC} = 0$: $3(-3) + 4(k + 1) = 0 \Rightarrow -9 + 4k + 4 = 0 \Rightarrow 4k - 5 = 0 \Rightarrow k = \frac{5}{4} = 1.25$. (c) When $k = 3$: $\vec{AC} = (-3; 4)$. Lengths: $|\vec{AB}| = \sqrt{3^2 + 4^2} = 5$, $|\vec{AC}| = \sqrt{(-3)^2 + 4^2} = 5$. Dot product: $\vec{AB} \cdot \vec{AC} = 3(-3) + 4(4) = -9 + 16 = 7$..

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the equation $2\cos 2x = 1 + \sin x$ for $0 \le x \le 2\pi$ radians, giving answers to 3 significant figures.

Substitute the double angle identity $\cos 2x = 1 - 2\sin^2 x$: $2(1 - 2\sin^2 x) = 1 + \sin x$ $2 - 4\sin^2 x = 1 + \sin x \Rightarrow 4\sin^2 x + \sin x - 1 = 0$ Apply quadratic formula for $\sin x$: $\sin x = \frac{-1 \pm \sqrt{1 - 4(4)(-1)}}{8} = \frac{-1 \pm \sqrt{17}}{8}$ Case 1: $\sin x = \frac{-1 + \sqrt{17}}{8} \approx 0.3904$: $x = 0.401$ rad, $\pi - 0.401 = 2.74$ rad Case 2: $\sin x = \frac{-1 - \sqrt{17}}{8} \approx -0.6404$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the binomial expansion of $\left(x^2 - \frac{2}{x}\right)^9$. (a) Write down the general term in terms of r. (b) Find the coefficient of x^3. (c) Find the term independent of x.

(a) General term: $T_{r+1} = \binom{9}{r}(x^2)^{9-r}\left(-\frac{2}{x}\right)^r = \binom{9}{r}(-2)^r x^{2(9-r)} x^{-r} = \binom{9}{r}(-2)^r x^{18-3r}$. (b) To find the coefficient of x^3, set the power of x equal to 3: $18 - 3r = 3 \Rightarrow 3r = 15 \Rightarrow r = 5$. Substitute $r = 5$: $T_6 = \binom{9}{5}(-2)^5 x^3 = 126 \times (-32)x^3 = -4032x^3$. The coefficient of x^3 is -4032. (c) The term independent of x has power 0: $18 - 3r = 0 \Rightarrow 3r = 18 \Rightarrow r = 6$. Substitute.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

Step 1: Write the general term T_{k+1}: $T_{k+1} = \binom{9}{k}(2x^2)^{9-k}\left(-\frac{1}{x}\right)^k = \binom{9}{k}2^{9-k}x^{2(9-k)}(-1)^k x^{-k} = \binom{9}{k}2^{9-k}(-1)^k x^{18-3k}$. Step 2: Find k for the term independent of x: Set exponent equal to 0: $18 - 3k = 0 \Rightarrow 3k = 18 \Rightarrow k = 6$. Step 3: Evaluate coefficient at $k = 6$: $\binom{9}{6} = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$..

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The curve C has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on C where the tangent is parallel to the y-axis (vertical).

Differentiate implicitly with respect to x: $3x^2 - 2\left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0$
$3x^2 - 2y + (2y - 2x)\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{2y - 3x^2}{2y - 2x}$ A vertical tangent occurs when the denominator is zero (and numerator non-zero): $2y - 2x = 0 \Rightarrow y = x$ Substitute $y = x$ into the original curve equation:
$x^3 - 2x(x) + x^2 = 4 \Rightarrow x^3 - x^2 - 4 = 0$ Test integer factors: $x = 2$ gives $8 - 4 - 4 = 0$. Factoring gives $(x - 2)(x^2 + x + 2) = 0$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

Since $\deg(\text{num}) = 3$ and $\deg(\text{denom}) = 2$ with $3 - 2 = 1$, a slant asymptote exists. Perform polynomial long division: $(2x^3 - 3x^2 + 5) \div (x^2 + 1) = (2x - 3) + \frac{-2x + 8}{x^2 + 1}$. The linear quotient gives the slant asymptote: $y = 2x - 3$. To find where the graph intersects this asymptote, set the remainder equal to zero: $\frac{-2x + 8}{x^2 + 1} = 0 \Rightarrow -2x + 8 = 0 \Rightarrow x = 4$. Substitute $x = 4$ into the asymptote equation:
$y = 2(4) - 3 = 5$. The curve intersects.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Practice 3.1: Solve the equation $2\cos 2x = 4\cos x - 3$ for $0 \le x \le 2\pi$, giving exact solutions.

1. Substitute the double angle identity $\cos 2x = 2\cos^2 x - 1$:
$2(2\cos^2 x - 1) = 4\cos x - 3 \Rightarrow 4\cos^2 x - 2 = 4\cos x - 3 \Rightarrow 4\cos^2 x - 4\cos x + 1 = 0$. 2. Recognize the perfect square:
$(2\cos x - 1)^2 = 0 \Rightarrow 2\cos x - 1 = 0 \Rightarrow \cos x = \frac{1}{2}$. 3. Reference angle: $\alpha = \arccos(1/2) = \frac{\pi}{3}$. Cosine is positive in Quadrants I and IV: In Q1: $x = \frac{\pi}{3}$. In Q4: $x = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$. Exact solutions.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

Substitute standard Cartesian-to-polar relations $x^2 + y^2 = r^2$, $x = r\cos\theta$, and $y = r\sin\theta$:
$(r^2)^2 = 4(r^2\cos^2\theta - r^2\sin^2\theta) \Rightarrow r^4 = 4r^2(\cos^2\theta - \sin^2\theta)$. Apply the double-angle identity
$\cos^2\theta - \sin^2\theta = \cos(2\theta)$: $r^4 = 4r^2\cos(2\theta)$. Divide through by r^2 (for $req0$): $r^2 = 4\cos(2\theta)$. This is a lemniscate (figure-eight curve) centered at the pole with loops oriented along the polar axis and maximum reach $r = 2$ at.

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

Factor numerator and denominator completely: $R(x) = \frac{(2x - 1)(x + 3)}{(x + 1)(x + 3)}$. The factor $(x + 3)$ cancels for $xeq -3$. Evaluate the simplified form $g(x) = \frac{2x - 1}{x + 1}$ at $x = -3$: $g(-3) = \frac{2(-3) - 1}{-3 + 1} = \frac{-7}{-2} = 3.5$. Key features:Hole: $(-3, 3.5)$Vertical asymptote: $x + 1 = 0 \Rightarrow x = -1$Horizontal asymptote: Equal degrees (2 and 2) $\Rightarrow y = 2/1 = 2$$x$-intercept: Numerator root $2x - 1 = 0 \Rightarrow (0.5, 0)$$y$-intercept.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 16.01 Permutations & Combinations

Solve the logarithmic equation $\log_3(x + 6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

Combine the logarithms on the left-hand side using the product rule: $\log_3[x(x + 6)] = 3$. Convert to exponential form: $x(x + 6) = 3^3 = 27 \Rightarrow x^2 + 6x - 27 = 0$. Factor the quadratic: $(x + 9)(x - 3) = 0 \Rightarrow x = -9$ or $x = 3$. Check the domain restrictions in the original equation:For $x = -9$: $\log_3(-9)$ is undefined, so $x = -9$ is an extraneous solution.For $x = 3$: $\log_3(9) + \log_3(3) = 2 + 1 = 3$. Valid ✓.Therefore, the unique solution is $x = 3$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 17.01 Discrete Random Variables

Practice 5.1: A curve has equation $y = x^3 - 3x^2 - 9x + 5$. (a) Find the coordinates of the local maximum and local minimum points. (b) Determine the coordinates of the point of inflection and find the equation of the tangent at the point of inflection.

(a) $y' = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$. Critical points at $x = -1$ and $x = 3$. $y'' = 6x - 6$. At $x = -1$: $y''(-1) = -120 \Rightarrow$ local minimum. $y(3) = 27 - 27 - 27 + 5 = -22$. Local min: $(3, -22)$. (b) Point of inflection: $y'' = 6x - 6 = 0 \Rightarrow x = 1$. $y(1) = 1 - 3 - 9 + 5 = -6$. Inflection point: $(1, -6)$. Slope at $x = 1$: $m = y'(1) = 3(1)^2 - 6(1) - 9 = -12$. Tangent line:

$y - (-6) = -12(x - 1) \Rightarrow y + 6 = -12x + 12 \Rightarrow y = -12x + 6$.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 18.01 Binomial Distribution

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

Step 1: Identify ellipse parameters: The vertices and foci lie on the x-axis, so the major axis is horizontal and centered at $(0, 0)$. Here semi-major axis $a = 5$ and focal distance $c = 3$. Step 2: Compute semi-minor axis b: For an ellipse: $b^2 = a^2 - c^2 = 5^2 - 3^2 = 25 - 9 = 16 \Rightarrow b = 4$. Step 3: Formulate equation and invariants: $\frac{x^2}{25} + \frac{y^2}{16} = 1$. Major axis length $2a = 10$, minor axis length $2b = 8$, and eccentricity $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 19.01 Normal Distribution

Practice AI.2: A company takes out a loan of 120,000 to purchase equipment at an annual interest rate of 6% compounded monthly, repayable by equal monthly payments over 10 years. (a) Find the amount of each monthly payment. (b) Calculate the total interest paid over the life of the loan.

(a) Use the annuity formula or TVM solver: Number of payments $N = 10 \times 12 = 120$. Monthly interest rate $r = \frac{0.06}{12} = 0.005$. $PV = PMT\left[\frac{1-(1+r)^{-N}}{r}\right] \Rightarrow 120,000 = PMT\left[\frac{1-(1.005)^{-120}}{0.005}\right]$. Compute discount factor: $1 - (1.005)^{-120} = 1 - 0.549633 = 0.450367$. Factor / 0.005 $= 90.07345$. x1,332.25. $(b) Total amount paid$: 120 \times 1332.25 = \159,870. Total interest: x159,870 - \$39,870.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Graphs & Transformations

The first term of a geometric sequence is $u_1 = 18$ and the common ratio is $r = \frac{1}{3}$. (a) Find the exact value of the 5th term u_5. (b) Calculate the sum of the first 6 terms, S_6. (c) State why the sum to infinity S_∞ exists, and find its exact value.

(a) $u_5 = u_1 r^4 = 18\left(\frac{1}{3}\right)^4 = 18\left(\frac{1}{81}\right) = \frac{18}{81} = \frac{2}{9}$. (b) $S_6 = \frac{u_1(1-r^6)}{1-r} = \frac{18(1-(1/3)^6)}{1-1/3} = \frac{18(1-1/729)}{2/3} = 27\left(\frac{728}{729}\right) = \frac{728}{27} \approx 26.96$. (c) Since $|r| = \left|\frac{1}{3}\right| < 1$, the terms converge to 0 and the infinite series converges. $S_\infty = \frac{u_1}{1-r} = \frac{18}{1-1/3} = \frac{18}{2/3} = 27$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Coordinate Geometry

Practice 2.1: The function $f(x) = 3 - \ln(2x + 1)$ is defined for $x > -0.5$. Find an expression for $f^{-1}(x)$, state its domain and range, and write down the coordinates of the intercept of $y = f^{-1}(x)$ with the y-axis.

1. Let $y = 3 - \ln(2x + 1)$. 2. Rearrange: $\ln(2x + 1) = 3 - y \Rightarrow 2x + 1 = e^{3-y} \Rightarrow x = \frac{e^{3-y} - 1}{2}$. 3. Swap variables: $f^{-1}(x) = \frac{e^{3-x} - 1}{2}$. 4. Domain of f^{-1} is the range of f, which is $\mathbb{R}$ (all real numbers). Range of f^{-1} is the domain of f, $(-0.5, \infty)$. 5. The y-intercept occurs at $x = 0$: $f^{-1}(0) = \frac{e^3 - 1}{2}$. Coordinates: $\left(0, \frac{e^3 - 1}{2}\right)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the simultaneous equations $y = 2x - 3$ and $x^2 + y^2 = 26$. Which pair of points represents the solutions?

a. $(1, -1)$ **and** $\left(\frac{17}{5}, \frac{19}{5}\right)$ ✓

b. $(3, 3)$ and $(-1, -5)$

c. $(5, 7)$ and $(1, -1)$

d. $(2, 1)$ and $(-2, -7)$

Substitute $y = 2x - 3$: $x^2 + (2x-3)^2 = 26 \Rightarrow 5x^2 - 12x + 9 = 26 \Rightarrow 5x^2 - 12x - 17 = 0 \Rightarrow (5x - 17)(x + 1) = 0$. So $x = -1 \Rightarrow y = -5$? Wait, check $x = -1$: $(-1)^2 + (-5)^2 = 26$! Yes, $(-1, -5)$ and $(17/5, 19/5)$. Let's verify choice A.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The line $x + y = 5$ intersects the hyperbola $xy = 6$. Find the coordinates of the two intersection points.

a. $(2, 3)$ **and** $(3, 2)$ ✓

b. $(1, 6)$ and $(6, 1)$

c. $(2, 3)$ and $(-2, -3)$

d. $(4, 1)$ and $(1, 4)$

$y = 5 - x \Rightarrow x(5 - x) = 6 \Rightarrow x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0$. Points are $(2, 3)$ and $(3, 2)$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the points of intersection of the line $2x - y = 1$ and the curve $y = x^2 - 4$.

a. $(3, 5)$ **and** $(-1, -3)$ ✓

b. $(2, 3)$ and $(-2, -5)$

c. $(4, 7)$ and $(1, 1)$

d. $(5, 9)$ and $(0, -1)$

$y = 2x - 1 \Rightarrow x^2 - 4 = 2x - 1 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$. For $x = 3$, $y = 5$; for $x = -1$, $y = -3$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the quadratic inequality $x^2 - 2x - 8 < 0$.

a. $-2 < x < 4$ ✓

b. $x < -2$ or $x > 4$

c. $-4 < x < 2$

d. $x < -4$ or $x > 2$

Factor: $(x - 4)(x + 2) < 0$. The parabola is below the x-axis between roots: $-2 < x < 4$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the set of values of x satisfying $2x^2 + 5x - 3 \geq 0$.

a. $x \leq -3$ **or** $x \geq \frac{1}{2}$ ✓

b. $-3 \leq x \leq \frac{1}{2}$

c. $x \leq -\frac{1}{2}$ or $x \geq 3$

d. $-1 \leq x \leq 3$

Factor: $(2x - 1)(x + 3) \geq 0$. Roots at $x = -3$ and $x = 1/2$. Product is non-negative outside roots: $x \leq -3$ or $x \geq 1/2$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the simultaneous inequalities $3x - 5 < 7$ and $x^2 - 9 \geq 0$.

a. $x \leq -3$ **or** $3 \leq x < 4$ ✓

b. $-3 \leq x \leq 3$

c. $x < 4$

d. $3 \leq x < 4$

$3x < 12 \Rightarrow x < 4$. $x^2 - 9 \geq 0 \Rightarrow x \leq -3$ or $x \geq 3$. Intersection gives $x \leq -3$ or $3 \leq x < 4$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what range of values of m does the line $y = mx$ not intersect the curve $y = x^2 + 9$?

a. $-6 < m < 6$ ✓

b. $m < -6$ or $m > 6$

c. $-3 < m < 3$

d. $m > 6$

Set $x^2 - mx + 9 = 0$. No intersection requires $\Delta < 0$: $m^2 - 36 < 0 \Rightarrow -6 < m < 6$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The perimeter of a rectangle is 28 cm and its area is at least 45 cm^2. What is the range of possible values for the width w ($w \leq l$)?

a. $5 \leq w \leq 7$ ✓

b. $4 \leq w \leq 9$

c. $3 \leq w \leq 7$

d. $w \geq 5$

$l + w = 14 \Rightarrow l = 14 - w$. Area $w(14 - w) \geq 45 \Rightarrow 14w - w^2 \geq 45 \Rightarrow w^2 - 14w + 45 \leq 0 \Rightarrow (w - 5)(w - 9) \leq 0 \Rightarrow 5 \leq w \leq 9$. Since $w \leq l$, width is $5 \leq w \leq 7$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the coordinates of the points where the line $y = 3x - 2$ is normal to the curve $y = x^2 - 4x + 3$.

a. $(1, 0)$ ✓

b. $(2, -1)$

c. $(0, -2)$

d. $(3, 0)$

Curve slope is $dy/dx = 2x - 4$. Normal slope must be $-1/(2x-4) = 3 \Rightarrow 2x - 4 = -1/3 \Rightarrow 2x = 11/3 \Rightarrow x = 11/6$.

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the range of values of k for which the line $y = kx - 1$ intersects the circle $x^2 + y^2 = 4$ in two distinct points.

a. $k^2 > -\frac{3}{4}$ **(all real k)** ✓

b. $k > \sqrt{3}$

c. $k < -\sqrt{3}$ or $k > \sqrt{3}$

d. $-1 < k < 1$

Substitute: $x^2 + (kx - 1)^2 = 4 \Rightarrow (1 + k^2)x^2 - 2kx - 3 = 0$.

$\Delta = 4k^2 - 4(1 + k^2)(-3) = 4k^2 + 12(1 + k^2) = 16k^2 + 12$. Since $16k^2 + 12 > 0$ for all real k, the line intersects the circle in two points for all real k.

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A company finds that its profit P (in thousands of dollars) satisfies $P(x) = -2x^2 + 28x - 80 \geq 10$, where x is the production level in hundreds of units. What is the valid production interval for x?

a. $5 \leq x \leq 9$ ✓

b. $4 \leq x \leq 10$

c. $3 \leq x \leq 11$

d. $6 \leq x \leq 8$

$-2x^2 + 28x - 90 \geq 0 \Rightarrow x^2 - 14x + 45 \leq 0 \Rightarrow (x - 5)(x - 9) \leq 0 \Rightarrow 5 \leq x \leq 9$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all values of x satisfying the rational inequality $\frac{x-3}{x+2} < 0$.

a. $-2 < x < 3$ ✓

b. $x < -2$ or $x > 3$

c. $x < 3$

d. $-3 < x < 2$

The quotient changes sign at $x = -2$ and $x = 3$. Testing intervals shows the quotient is negative strictly when $-2 < x < 3$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Relative to a fixed origin O, the lines L_1 and L_2 have vector equations $L_1 : \mathbf{r} = (1; 0; -1) + \lambda(1; 2; 2), \quad L_2 : \mathbf{r} = (2; -4; 7) + \mu(2; 1; -2)$. The unique points P on L_1 and Q on L_2 are such that the line segment PQ is perpendicular to both L_1 and L_2. A point R lies on L_1 such that the area of triangle PQR is 18. Given that the z-coordinate of R is positive, what is the exact distance from the origin O to the midpoint of the line segment QR?

a. $3\sqrt{5}$ ✓

b. $\sqrt{13}$

c. $6\sqrt{2}$

d. $\sqrt{77}$

1. **Determine the positions of P and Q:** Let P on L_1 have coordinates $(1 + \lambda, 2\lambda, -1 + 2\lambda)$ and Q on L_2 have coordinates $(2 + 2\mu, -4 + \mu, 7 - 2\mu)$. The displacement vector $\vec{PQ}$ is: $\vec{PQ} = ((2 + 2\mu) - (1 + \lambda); (-4 + \mu) - 2\lambda; (7 - 2\mu) - (-1 + 2\lambda)) = (1 + 2\mu - \lambda; -4 + \mu - 2\lambda; 8 - 2\mu - 2\lambda)$. Since $PQ \perp L_1$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A curve passes through the point $\left(1, \frac{1}{2}\right)$ and satisfies the differential equation $\frac{dy}{dx} = \frac{y^2(1 - \ln x)}{x^2}$ for $x > 0$. Determine the exact coordinates of the stationary point of the curve and its nature.

a. $\left(e, \frac{e}{2e-1}\right)$**, which is a local maximum** ✓

b. $\left(e, \frac{1}{2e-1}\right)$, which is a local maximum

c. $\left(e, \frac{e}{2e+1}\right)$, which is a local minimum

d. $\left(\frac{1}{e}, \frac{1}{2e+1}\right)$, which is a local minimum

1. **Separate the variables:** $\frac{1}{y^2}\,dy = \frac{1 - \ln x}{x^2}\,dx$ 2. **Integrate both sides:** The left-hand side integrates to: $\int y^{-2}\,dy = -\frac{1}{y}$ For the right-hand side, split into two integrals: $\int \frac{1 - \ln x}{x^2}\,dx = \int x^{-2}\,dx - \int x^{-2} \ln x\,dx$ We have $\int x^{-2}\,dx = -\frac{1}{x}$. Using integration by parts for $\int x^{-2} \ln x\,dx$ with $u = \ln x$ and $v' = x^{-2}$.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A curve satisfies the differential equation $\frac{dy}{dx} = \frac{k-y}{2x+1}$ for $x > -\frac{1}{2}$, where k is a real constant. The curve passes through the points $(0, 8)$ and $(4, 2)$. Find the value of k.

a. -1 ✓

b. 11

c. $\frac{5}{4}$

d. 5

To find the constant k, we solve the differential equation by separation of variables: $\int \frac{1}{k-y}\,dy = \int \frac{1}{2x+1}\,dx$ Integrating both sides (paying close attention to the chain rule): $-\ln|k-y| = \frac{1}{2}\ln(2x+1) + C$ Multiplying by -1 and exponentiating: $\ln|k-y| = -\frac{1}{2}\ln(2x+1) - C = \ln\left((2x+1)^{-1/2}\right) - C$
$|k-y| = \frac{A}{\sqrt{2x+1}}$ where $A = e^{-C} > 0$ Removing the absolute value gives th.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the particular solution of the differential equation $\frac{dy}{dx} = 2xe^{-y}$ given that $y = 0$ when $x = 0$.

a. $y = \ln(x^2 + 1)$ ✓

b. $y = \ln(x^2)$

c. $y = \ln(2x^2 + 1)$

d. $y = x^2$

To solve the differential equation, we use separation of variables: 1. **Separate the variables**: $e^y\,dy = 2x\,dx$ 2. **Integrate both sides**: $\int e^y\,dy = \int 2x\,dx$ $e^y = x^2 + C$ 3. **Apply the initial condition** $y = 0$ when $x = 0$: $e^0 = 0^2 + C \Rightarrow 1 = C$ *(A frequent error is assuming $C = 0$ because the boundary values are zero, which leads to the incorrect solution $y = \ln(x^2)$.)* 4. **Express y explicitly in terms of x**.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.

a. Consistent and independent (one solution)

b. **Consistent and dependent (infinitely many solutions)** ✓

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 19.01 Normal Distribution

Find the projection of $\mathbf{u} = \langle 3, 2 \rangle$ onto $\mathbf{v} = \langle 2, 2 \rangle$.

a. $\left\langle \frac{3}{2}, \frac{5}{2} \right\rangle$

b. $\left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$ ✓

c. $\left\langle \frac{7}{2}, \frac{5}{2} \right\rangle$

d. $\frac{5}{4}$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} = \frac{10}{8}\langle 2, 2 \rangle = \left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Graphs & Transformations

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 3$. What is the value of x?

a. **12** ✓

b. 3

c. 6

d. 14

From the first equation: $\log_2(x/y) = 2 \Rightarrow x/y = 2^2 = 4 \Rightarrow x = 4y$. Substitute into second: $4y - 3y = 3 \Rightarrow y = 3$. Thus $x = 4(3) = 12$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Coordinate Geometry

Find the angle between $\mathbf{u} = \langle 2, 1 \rangle$ and $\mathbf{v} = \langle 4, 2 \rangle$ (to one decimal place).

a. **0.0°** ✓

b. 20.0°

c. 180.0°

d. 10.0°

$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|} = \frac{10}{2.24 \times 4.47} \approx 1.0$, so $\theta \approx 0.0°$.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Trigonometry

Find the unit vector in the direction of $\mathbf{v} = \langle -8, 15 \rangle$.

a. $\left\langle \frac{9}{17}, \frac{15}{17} \right\rangle$

b. $\left\langle \frac{15}{17}, -\frac{8}{17} \right\rangle$

c. $\left\langle -\frac{8}{17}, \frac{15}{17} \right\rangle$ ✓

d. $\langle -8, 15 \rangle$

$|\mathbf{v}| = 17$. Unit vector $= \frac{\mathbf{v}}{|\mathbf{v}|} = \left\langle \frac{-8}{17}, \frac{15}{17} \right\rangle$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Differentiation & Integration

Which differential equation correctly models the statement: "The rate of decrease of the temperature T of an object is proportional to the difference between T and the surrounding temperature T_e."?

a. $\frac{dP}{dt} = kP$

b. $\frac{dT}{dt} = -k(T - T_e)$ ✓

c. $\frac{dN}{dt} = kN(S - N)$

d. $\frac{dm}{dt} = -km$

The correct model is $\frac{dT}{dt} = -k(T - T_e)$, where $k > 0$ is a constant of proportionality.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Statistics & Probability

For the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$, what are the equations of the asymptotes?

a. $y = \pm\frac{3}{2}x$ ✓

b. $y = \pm\frac{2}{3}x$

c. $y = \pm 6x$

d. $y = \pm\frac{5}{2}x$

Asymptotes of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $y = \pm\frac{b}{a}x = \pm\frac{3}{2}x$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Functions: Composite & Inverse

Which differential equation correctly models the statement: "The rate of increase of the number of infected people N is proportional to the product of N and the number of remaining susceptible people $(S - N)$, for fixed population S."?

a. $\frac{dm}{dt} = -km$

b. $\frac{dV}{dt} = kr^2$

c. $\frac{dT}{dt} = -k(T - T_e)$

d. $\frac{dN}{dt} = kN(S - N)$ ✓

The correct model is $\frac{dN}{dt} = kN(S - N)$, where $k > 0$ is a constant of proportionality.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Circular Measure

Which differential equation correctly models the statement: "The rate of increase of the volume V of a balloon is proportional to the square of its radius r."?

a. $\frac{dT}{dt} = -k(T - T_e)$

b. $\frac{dm}{dt} = -km$

c. $\frac{dV}{dt} = kr^2$ ✓

d. $\frac{dP}{dt} = kP$

The correct model is $\frac{dV}{dt} = kr^2$, where $k > 0$ is a constant of proportionality.

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